

Efficient Error Detection and Correction code based on Improved Redundant Matrix Code (IRMC) Algorithm

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Abstract- Mobile way communication over the Internet requires error detection and correction. The detection of error at the data link layer can be controlled by network handling techniques. Data traffic, signaling and control will be carried out by two channels of network connection. Error detection will be done for traffic channels of the information bits, encoded correction code together forms the transmitted data stream. Currently, the error is tolerated by the SRAM memory frame, so Single Cell Upsets (SCU) or Multiple Cell Upsets (MCU) can occur. One of the main reasons for the MCU is astronomical radiation. The proposed technique is called Improved Redundant Matrix Code (IRMC) to protect memories from the detuning of many components, is used to detect and troubleshoot storage devices. IRMC's detection rate and adjustments contrast with other rescue codes. Space, power, and defer overhead is only. Results should prove that IRMC has higher quality and higher correctability for MCU tuned for the lower power and delay. Xilinx 14.7 is the tool used for the process and using Verilog HDL, programming of the algorithm is performed.

Keywords— Multiple error detection ,Error Detection Correction Code (EDCC), Improved Redundant Matrix Code (IRMC), Multiple Cell Upsets (MCU), Multiple error correction Memory.

I. INTRODUCTION

Systems for data transportation, communication provide services to consumers. Consumers can quickly assemble data for these services through parallel transmission. During transmission, all data streams are sent through different communication channels [1]. A direct correlation is established between the number of active lines and the number of bits transmitted. For a clock pulse group of bits is sent over the data lines. Practically speaking, parallel transmission requires "N" communication channels to work. Contrary to popular belief, sequential transmission requires only one communication channel, which reduces the total cost compared to serial transmission [2]. Memories will be solid only if the information stored inside is error-free. If there is an error, you will know at once.

Communication difficulties that arise during strategy implementation are caused by transmission errors. Soft errors are those caused by external noise or radiation rather than design or manufacturing defects [3-4].

When using a non-random code, the encoded message is modified to have a certain number of bits that are less likely. Another grouping is a type of systematic module, in which the transmitter sends unique data with a predetermined number of parity bits, such as check bits derived from information bits [5]. When only error detection is required, the beneficiary uses the same computation to compare the received information bits with that of the check bits. In the transmission process error will occurred when attribute do not match. Typically, error-correcting codes are used for lower-layer communication, equivalent to reliable limits in media such as RAM, CDs, hard disk and DVDs. [6-8].

The main purpose of using IRMC is to identify and repair defects in memory devices[15-16]. The range, power, and delay of other correction codes are comparable to IRMC's detection and correction speed [9]. The Improved redundancy matrix code (IRMC) algorithm is a coding technique used in Large Scale Integration (VLSI) technology to improve the reliability of digital circuits. This is a type of error correction code (ECC) that allows you to detect and correct errors that may occur during data transfer or storage [10].

II. IMPROVED REDUNDANT MATRIX CODE ALGORITHM

In the IRMC algorithm, a redundant bit matrix is added to the original data to generate a new code word . The identification of the bit error and correction can be done by designing proper redundant bit matrix. correction of the The IRMC algorithm is particularly useful in VLSI technology where the circuit density is very high, and the possibility of errors is high. The use of IRM code algorithm in VLSI can improve circuit reliability, reduce the risk of

data corruption, and improve overall system performance [11].

In addition, the IRMC algorithm is also used in other fields such as telecommunications, data storage, and satellite communications. This is a widely accepted method of error correction, and its use is expected to increase as technology advances. In the proposed work, the IRMC is executed for error detection and correction of 32-bit, 64-bit input codes using Verilog in Xilinx tool in the below mentioned.

Taking the Original 32-bit/ 64-bit as input for Encoder of IRMC algorithm. Generating Horizontal Redundant bits using AXL calculation and Vertical Redundant bits using Logical Organization of Original input in Encoder region. Taking Erroneous 32-bit/64-bit as input for decoder part. Performing step (ii) for the erroneous input to get respective horizontal and vertical redundant bits. Generating Syndrome value to detect and correct the error, if error exists. Using the generated syndrome value, the error can be detected and corrected, that resulting into the erroneous input bit will be regenerated, which should be same as that of original input bit. Generating Simulation results & Design report using Xilinx tool. Delay and Power Analysis of IRMC Algorithm. Comparing the resulted IRMC algorithm delay and power with different algorithms[12-13].

III. ENCODER OF IRMC ALGORITHM

The Enhanced Redundancy Matrix Coding algorithm that can detect and correct multiple errors in a message. It works by creating a parity check matrix that is used to detect errors in received messages. The same matrix is created by adding additional rows and columns to the original message matrix [1]. These additional rows and columns contain parity bits used to check the integrity of the

message. The step by step process is performed, to encode a message using enhanced redundancy matrix encoding algorithm[14-15].

Message is converted into binary code. Binary messages are divided into blocks of equal size. The redundancy matrix is created by adding additional rows and columns to the original message matrix. These additional rows and columns contain parity bits used. The matrix with message, the redundancy matrix are multiplied together to produce the encoded message. Coded messages are sent or saved[16-17].

By using PCM it is very easy to identify the adult location while transmitting the data. If the error is a single error, it can be corrected using the parity bit. If some errors are detected, the message may need to be sent. Overall, the error detection and correction code encoder using the improved or enhanced redundancy matrix code algorithm shows a suitable method for detecting and correcting errors in messages. It is widely used in various applications such as communication systems, data storage systems, and digital media [18].

Consider, an encoder was introduced that registers repeating bits using logic arrays and XOR inputs. Traditional parts are added to identifying the defects. It is clearly indicated that from the fig.1 horizontal bits and vertical bits are represented as $(H_{15}-H_0)$ and $(V_{15}-V_0)$ respectively for generated the 32-bit input $(B_{31}-B_0)$.

The algorithm in the IRMC encoder starts with the same matrix. Choosing the logic conditions to calculate each equation and syndrome is not difficult when the equation matrix is chosen. For example, the AXL calculation uses symbols S_0 ($B_3B_2B_1B_0$) and S_2 ($B_{11}B_{10}B_9B_8$) to obtain the extra bits $[H_3H_2H_1H_0]$, which is shown in Fig.2.

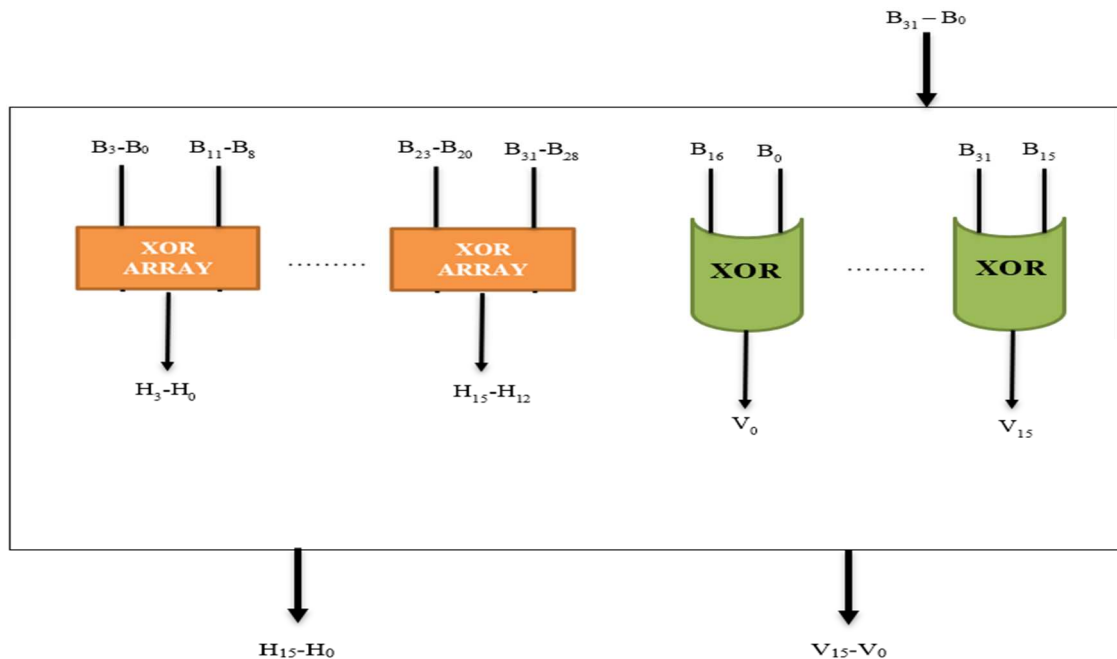


Fig. 1. Block Diagram of IRMC Encoder.

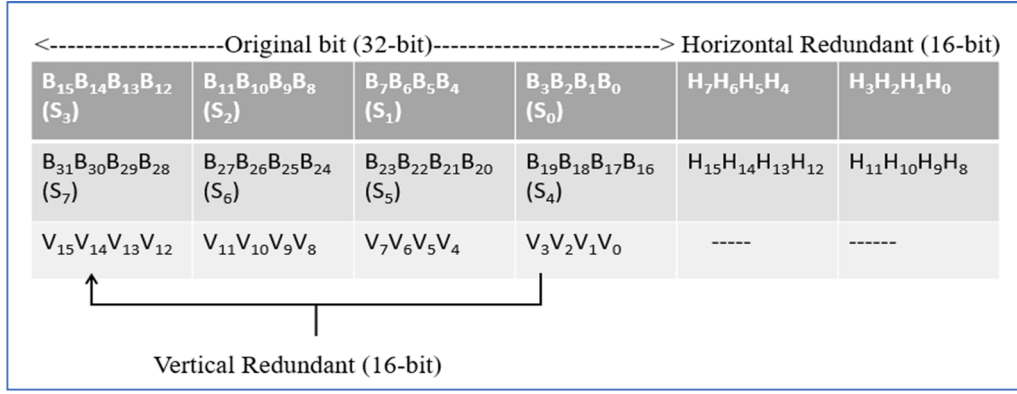


Fig. 2. Logical Organization of Original input and redundant bits.

AXL the horizontal and vertical redundant bits will be computed as shown :

Horizontal Bits generation using AXL calculation,

$$B_{11}B_{10}B_9B_8 \wedge B_3B_2B_1B_0 = H_3H_2H_1H_0 \quad (1)$$

$$B_{15}B_{14}B_{13}B_{12} \wedge B_7B_6B_5B_4 = H_7H_6H_5H_4 \quad (2)$$

Vertical Bits generation

$$V_n = B_n \wedge B_{n+16} \quad (3)$$

where n=number of bits

IV. DECODER OF IRMC ALGORITHM

A decoder for error detection and correction using an improved redundancy matrix coding algorithm is a system designed to encode messages that may contain errors [13]. The encoder uses the parity check matrix generated during the encoding process to identify and correcting the errors in the received message. The encoding process begins with the received message, which is usually sent or stored as a binary string. The first step in the encoding process is to perform a similarity check on the received message using the similarity matrix created in the encoding process.

The equation matrix is a square matrix derived from the original message and the redundancy matrix. This is used to check the involvement of the message by comparing the received message with encoded message. when the error is presented that location and type of error can be predicted.If a bit error is detected, the decoder can correct the error by inverting the bit. When more number of errors are identified, the decoder may request that the message be retransmitted or use more advanced error correction techniques

message. Detecting and correcting errors in messages provides an efficient and effective way to ensure accurate and reliable transmission. The schematic diagram of decoder is shown in fig.3.

About the terms used in Decoder,

B → Original Input (consider 32-bit)

H → Horizontal redundant bits (16-bit)

V → Vertical redundant bits (16-bit)

Be → Erroneous Input (32-bit)

He → Horizontal redundant bits for erroneous input (16-bit)

Ve → Vertical redundant bits for erroneous input (16-bit)

Sp → Syndrome value (16-bit)

Spm → Updated syndrome value (32-bit)

C → Final modified output with no error bits (32-bit)

The equations used in decoder are,

i. Equation used for error identifier (YES/NO),

if $H \wedge H^e = 0$ (3); no error exists (NO)

if $H \wedge H^e \neq 0$ (4); error exists (YES)

ii. Equation used to generate syndrome,

$$S_p = V \wedge V^e \quad (5)$$

iii. Equation used to result the error less output code,

$$C = S_{pm} \wedge B^e \quad (6)$$

To calculate the syndrome, the decoder performs mathematical operations on the received data with redundant bits, usually by multiplying by a predetermined matrix. This process generates a set of output values that can be compared against a predefined set of expected values to detect any deviations or errors. If the syndrome indicates that there is an error, the decoder can use several methods to correct the error. The syndrome is an important part of the error detection and correction process in the code using the improved coding redundancy matrix algorithm. By calculating and analysing syndromes, the decoder can identify and correct errors that may occur during transmission, ensuring that the data received is accurate and reliable.

V. WORKING OF IRMC ALGORITHM FOR 32-BIT INPUT

Input Parameters & Flow of process to identify and rectify error bits, Consider,

B = 1111 0101 1010 1111 1111 0110 1010 1100 (31st bit to 0th bit)

Generating horizontal & vertical parameters for original input B

H = 0101 1010 0101 1010

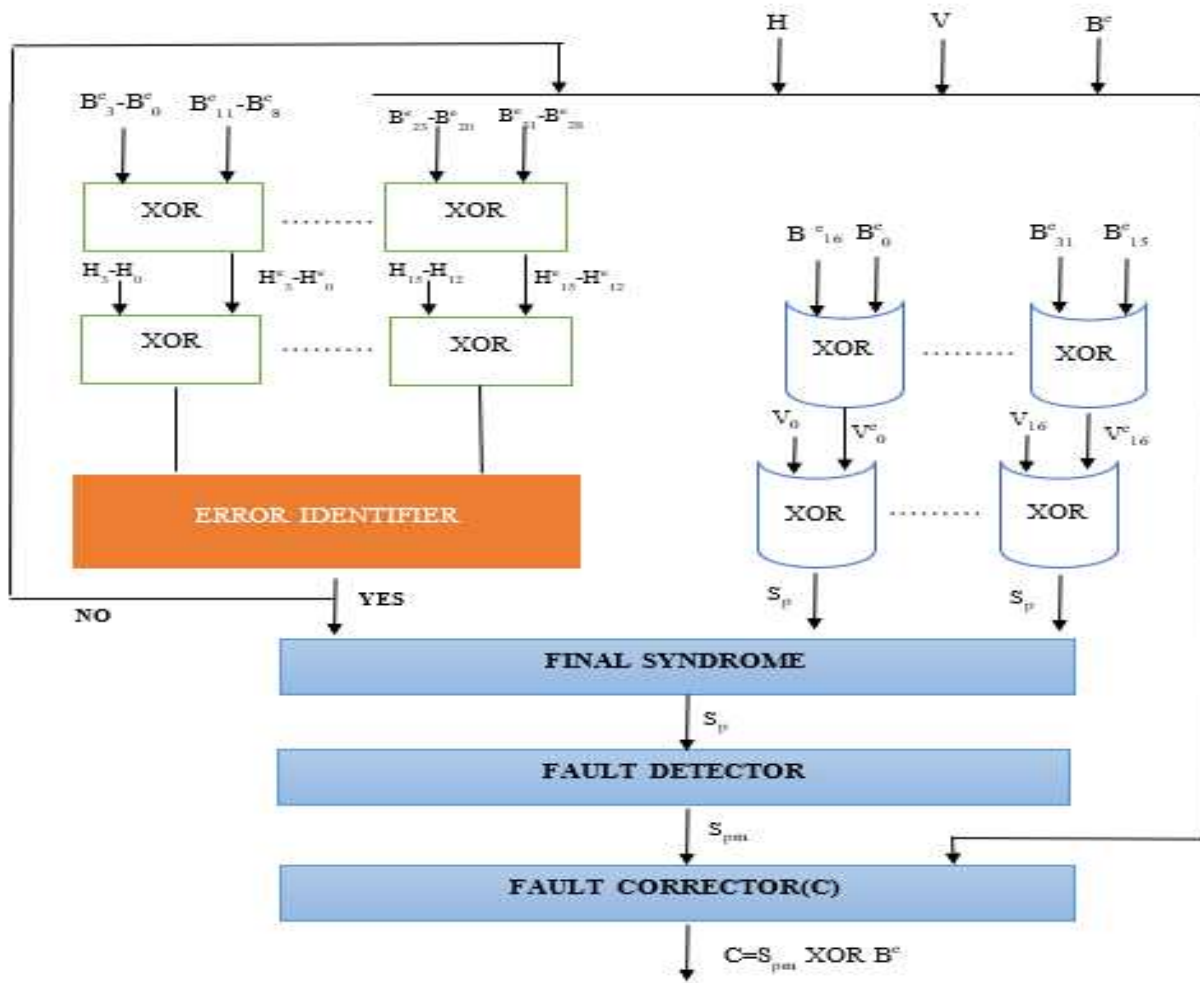


Fig. 3. Block Diagram of IRMC Decoder

		31 st bit		15 th bit													0 th bit
B		1	1	1	1	0	1	1	0	1	0	1	0	1	1	0	0
V		1	1	1	1	0	1	0	1	1	0	1	0	1	1	1	1
B^e		1	1	0	1	0	1	0	0	1	1	1	1	1	1	0	0
V^e		1	1	1	1	1	1	0	1	1	0	1	0	1	1	0	1
S_p		0	0	1	0	1	0	1	0	1	1	0	1	0	0	1	0
C		1	1	1	1	0	1	1	0	1	0	1	0	1	1	0	0
		1	1	1	1	0	1	0	1	1	0	1	0	1	1	1	1

Fig. 4. Correction of Error Bits

$V = 0000\ 0011\ 0000\ 0011$

Consider, Erroneous input,

$B^e = 1111\ 1101\ 1010\ 1101\ 1101\ 0100\ 0111\ 1100$

Generating horizontal & vertical parameters for B^e

$H^e = 0101\ 0000\ 1010\ 1000$

$V^e = 0010\ 1001\ 1101\ 0001$

After obtaining H , H^e , V and V^e

$H \wedge H^e = 0000\ 1010\ 1111\ 0010 (\neq 0)$

Since, error exists, we need to calculate Syndrome (S_p).

S_{pm} is simply generated by writing 16-bit S_p two times in a row.

Using S_{pm} , we rectify the error to obtain errorless output using $C = B^e \wedge S_{pm}$.

Hence, $C = 1111\ 0101\ 1010\ 1111\ 1111\ 0110\ 1010\ 1100$ (31st to 0th bit).

Fig.4 shows how the error bits are getting corrected using 2D forms, and also describes what are the errors located in the erroneous input to understand how the process is getting executed. Syndrome is S_p , is calculated by XORing($\wedge$) V and V^e .

VI. RESULTS

1. Implementation of IRMC Algorithm using Verilog HDL language for 32-bit input.

Simulation results, Design report & timing details of 32-bit encoded & decoded result are provided in the figures 5 shown below.

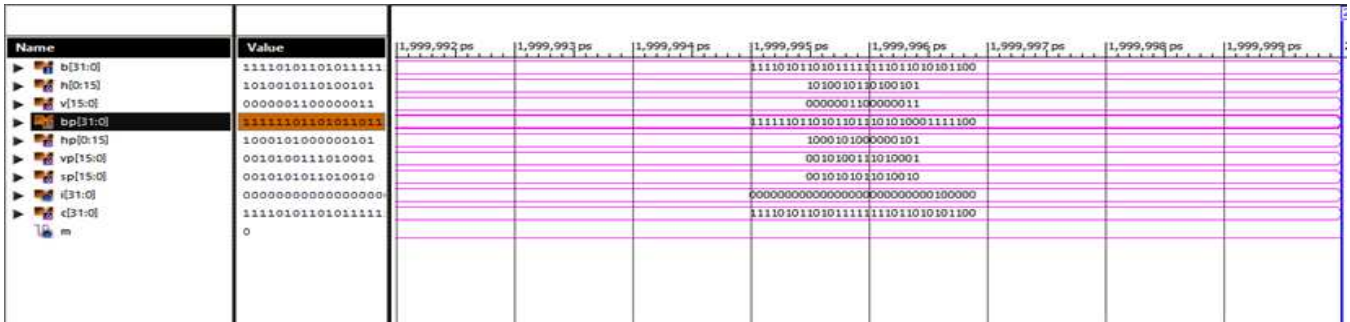


Fig. 5. Simulation result of 32-bit input.

2. Implementation of IRMC Algorithm for 64-bit input

It also shows whether the erroneous 32-bit is getting equal to original 32-bit after getting decoded.

Simulation results, Design report & timing details of 64-bit encoded & decoded result are provided in the coming figures:

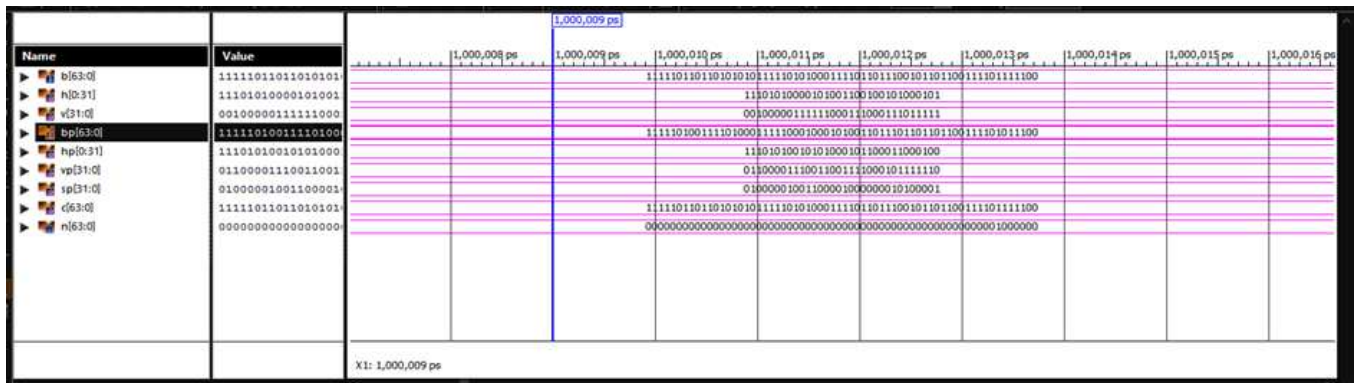


Fig. 6. Simulation result of 64-bit input.

3 FPGA Implementation of IRMC Algorithm for 16-bit input

The FPGA kit used for implementation is Artix-7 XC7A100t board is shown in fig.7. A 16-bit input was taken as original input and another 16-bit input was taken as erroneous input. To take these inputs from FPGA board an array was considered which has 4 rows that can store 8-bits per each row. To normalize the clock, a clock divider is used to give input for a specific time lapse. In the similar manner, the 16-bit output also will be displayed in 2 specific time frames as 8-bits at a time. Based on application requirements PPA will be optimized. The error finding and correction will be made easy with the reconfigurable feature of FPGA..



Fig. 7. FPGA implementation of IRMC.

VI. COMPARISON OF IRMC WITH STATE OF ART EXISTING ALGORITHMS

If we apply 32-bit data as input to the module, the delay value is 5.937ns and power is 38.903mW. The algorithms like DMC, BMC, OLS etc are similar to IRMC method. The simulation results of power and delay are compared with the above said methods. Comparison table for delay & power for 32-bit input is shown in Table.1

Table 1 : Delay and Power comparison between IRMC (proposed) and other existing algorithms

<i>Methods</i>	<i>Delay (ns)</i>	<i>Power (mW)</i>
IRMC (proposed)	5.937ns	38.903mW
Decimal Matrix Code[9]	17.587ns	121mW
Bit-Manipulation Algorithm[10]	6.065ns	327.3mW
Reed-Muller Code[1]	55ns	264.8mW

VII. CONCLUSION & FUTURE SCOPE

An Improved Redundancy Matrix code algorithm provides a reliable and efficient approach to error detection and correction in VLSI. With the increasing demand for suitable and reliable digital communication systems, these algorithms and systems plays an important role in improving the PPA performance, reliability of later upcoming future systems. Integration with technologies such as 5G, optimization of low-power devices, and implementation. in various hardware electronic systems etc.

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